

Requirements Analysis

CardioCare Quest

Version 2.0

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For the Client:_____

For the Team:_____

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Introduction

Context

Hypertension is estimated to affect **1.4 billion adults worldwide**, fueling a massive Mobile Health industry currently valued at over **\$70 billion, comprising** thousands of tracking apps and wearable technologies. Despite this overwhelming availability of modern medicine, only around **25%** of that population has their condition under control. It's clear that this industry suffers from a "homogenization" problem: tools are being built for a generalized Western culture, failing to connect with large portions of the global population who fall outside of these assumptions.

This disparity is most prominent within Indigenous communities, facing hypertension prevalence rates **10-20% higher than in non-Indigenous groups**¹. These issues emerge as a direct manifestation of systemic technological and healthcare inequalities alongside a severe cultural mismatch in medical interventions. When tools designed to save lives fail to speak the language of the people they serve, the result is a widening gap in health equity and a preventable loss of life.

The Problem

Current hypertension management tools rely on clinical tracking apps and pharmaceutical regimens that feel isolating and transactional. These solutions adopt a one-size-fits-all approach that promotes a negative experience between patients and their own health. Direct problem deficiencies are as follows:

- **Cultural Mismatch:** Standard medical models often ignore cultural backgrounds, resulting in low engagement and a profound lack of trust within Indigenous communities.
- **Technological Barriers:** Existing creative tools are too code-heavy and complex for non-technical users to design their own solutions.
- **Isolation:** Current medical apps promote a data-driven relationship between patients that ignores the social and cultural dimensions of health.

Our Client

Dr. Jared Duval², director of the **Playful Health Technology (PHT) Lab**³, primarily studies human-computer interaction technologies in order to enhance and maintain health within underserved populations. His work has led to a series of game technologies dedicated to

¹ [See Reference](#)

² [See Reference](#)

³ [See Reference](#)

dismantling these clinical barriers by engaging users in medically relevant forms of play. As of now, the goal of the Lab and this project is to address those hypertension disparities within the Indigenous communities of Arizona and Texas.

The Lab has previously defined a current solution using Twine, a game creation tool, allowing members of the community to create and import their own narratives to boost engagement and trust, and engage in a gamified experience while still tracking hypertension data. This idea alone, however, is inadequate because such tools can be too complex or code-heavy for non-technical users and still require game concepts to be fleshed out. It creates a fragmented system where the technology itself, much like existing medical apps, acts as a barrier rather than a bridge, preventing the exchange needed to make culturally grounded medical interventions successful.

Solution Vision

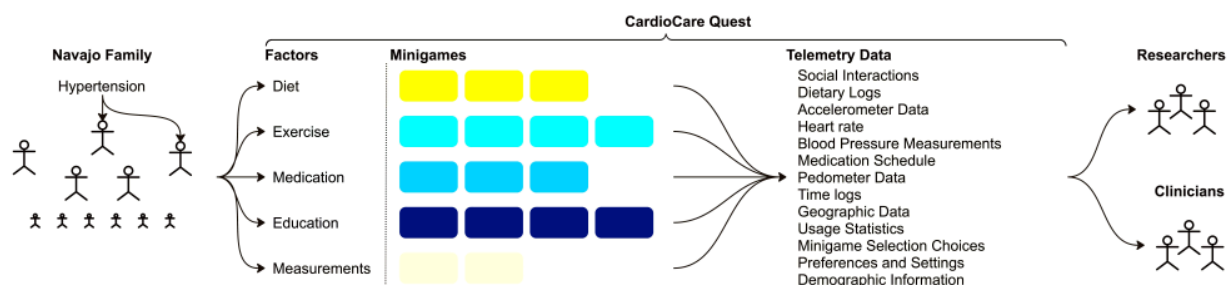
In response, we are collaborating with the PHT Lab to develop **CardioCare Quest**, a mobile health gaming environment meant to replace standard health tracking applications with a culturally grounded form of play within the Navajo community. This app is a platform that allows users to track hypertension data by engaging in Twine narratives that reflect the community's values and traditions.

Our role in this project is split into two core technical and creative responsibilities:

- The Unified App Shell:** We are designing a seamless experience that allows users to navigate between the main app and various sub-games with ease. This shell will be based on the PHT Lab's other application, *NetGauge*, to ensure a consistent visual style between projects.
- The Exemplar Games:** We will be creating 5 specialized Twine games inspired by workshops conducted within the community, each focused on collecting specific hypertension metrics for Dieting, Blood Pressure, Medicine Adherence, Exercise, and providing Education. These games will serve as high-fidelity examples, providing a blueprint for how future community-generated content can utilize storytelling as a form of medical data collection. This also includes extensive documentation and organization within our games to ensure non-technical users are able to produce their own Twine narratives.

Data Usage & Transformation

This system utilizes both qualitative and quantitative narrative choices within each game to collect self-reported HBP data from users and transform it into a list of telemetry data for clinicians and researchers. This data is stored privately to maintain user privacy, but can be displayed within the app to map progress over time and compete for leaderboard positions within the community.



By integrating these features, CardioCare Quest hopes to build a strong base for the future of medical data collection, whilst also having broader implications in the world of culturally responsive digital health innovations.

With the solution vision established, this document will now move into discussing requirements aimed at targeting the defined problem deficiencies while retaining full functionality of the features defined by the client.

Project Requirements

Domain-Level Requirements

The requirements listed below are viewed as the overarching framework for each individual game to follow. While the specific games handle different medical tasks, these requirements ensure the entire platform feels cohesive.

1. **De-Clinicalized Health Education:** This system aims to replace a data-centric design with an engaging, narrative-driven environment to provide information about hypertension management.
2. **Relational Design:** This system must respect cultural barriers and Indigenous epistemologies and provide a relational experience between users and their interactions within each game.
3. **Community-Led Content Ownership:** This platform must be designed for non-technical community members to audit, edit, and create their own content.
4. **Accessibility:** This system must be accessible between different forms of mobile hardware and to a wide variety of users to ensure high performance and availability for different users.
5. **Medical Data Handoff:** This system must reliably collect patient-generated data for health tracking.

Now that the domain rules have been defined, the following sections will break down into the Functional and Non-Functional requirements for each of the five games while applying the above philosophies to their unique sets of mechanics.

Twine Game 1: Blood Pressure Tracking

This twine game will be focused on recording the user's blood pressure. According to UCSF Health⁴, a patient should have been resting for at least 5 minutes prior to the test. Furthermore, a patient's blood pressure should only be measured when not under stress. Due to this, the goal of this game is largely to give the user something to do while resting to achieve accurate results, and remain relaxed while doing so. To achieve this, the "Blood Pressure Game" will itself be a host for many short minigames, each of which can be completed in at least 5 minutes.

⁴ [See Reference](#)

Functional Requirements

1. **Syncing Data with Clinical Instruments:**
 - 1.1. **Live Data:** When users have their blood pressure measured, the measurement should be automatically updated to the App from the clinical tool.
 - 1.2. **Timing:** The game should display to the user how much time has passed, allowing them to realize when the necessary time has been spent to accurately measure their blood pressure.
2. **Multitude of Minigames:**
 - 2.1. **Word Search:** A Simple word search game providing multiple options and categories for users to choose.
 - 2.2. **Free-Draw:** A grid-based drawing game, used for Users who just want to draw.
 - 2.3. **Cryptograms:** Text-unscrambling Game to reveal a famous phrase or quote. For users looking for a brain teaser.
 - 2.4. **Mazes:** A grid-based maze that users can navigate through.
 - 2.5. **Crossword:** A grid-based text game where clues are given to fill vertical and horizontal text boxes to solve the puzzle.
3. **Relaxing Content:**
 - 3.1. **No Time Limits:** No game should be restricted in time to prevent a sense of urgency and ensure a relaxed environment.
 - 3.2. **Auditory Limits:** Audio should be confined to calming sound bytes or relaxing music loops.
4. **User Control**
 - 4.1. **Theme control:** The color scheme and audio should be able to be controlled by the user so users can tailor their experience for themselves.
 - 4.2. **Game-Favoriting:** If a user enjoys a game enough, they should be able to add the game to their personal favorites, ensuring the game remains at the top of the game list.

Non-Functional Requirements

1. **Performance**
 - 1.1. **UI Latency:** All swaps between current minigames must take place within 100ms.
 - 1.2. **Responsive Gameplay:** All user input to the minigames must be recognized within 100ms, and be accompanied by either a visual or auditory cue, or both, to improve responsiveness.
2. **Accessibility:**
 - 2.1. **Accessible Mini-games:** Mini-games must be accompanied by 150 words or fewer instructions.
 - 2.2. **Text-To-Speech Instructions:** All text-to-speech options will be paired with instructions for users.

Twine Game 2: Education Game

Functional Requirements:

1. **User and Developer Created Trivia Games**
 - 1.1. **Predefined Categories:** Developers provide pre-created trivia categories that users can select from.
 - 1.2. **Personal Domains of Knowledge:** Players can create custom trivia categories based on personal, cultural, or contextual knowledge.
 - 1.3. **Saved User-defined Knowledge:** Store user-generated categories and question sets across sessions.
 - 1.4. **Use Case:** A user creates a category about traditional foods, adds questions, and plays with family members.
2. **Jeopardy-Style Gameplay System**
 - 2.1. **Point-grid Gameplay System:** A grid-based interface display with categorized questions and associated point values.
 - 2.2. **Question and answer flow:** Players select questions and submit answers through an input interface.
 - 2.3. **Category-based navigation:** Clearly labeled categories accessible from the main interface.
3. **Reusable Categories Across Sessions**
 - 3.1. **Player progress shall persist across sessions:** Player progress shall be automatically saved across sessions and application restarts.
 - 3.2. **User-created categories shall remain editable over time:** Players can modify, rename, or delete previously created categories at any time.
 - 3.3. **Categories reusable across sessions or games:** The system shall allow for categories to be reused across different gameplay sessions.
 - 3.4. **Use Case:** A user creates a category about traditional animals, closes the app, and sees the same category listed upon relaunch.

Non-Functional Requirements:

1. **Performance & Reliability**
 - 1.1. **Latency:** All user interactions shall respond within 100 ms under normal operating conditions
2. **User Friendly UI**
 - 2.1. **User reaction time and speed with familiar layout:** UI Population and question select should take place within 2 seconds on average.
 - 2.2. **Cognitive engagement through trace analysis:** Track time spent per question and user answer accuracy for performance analysis.
3. **Inviting & accessible**

- 3.1. **Inclusive participation model:** New users will be able to create and play a trivia game within 3 minutes without prior technical experience.
 - 3.2. **Emotional and Cultural safety:** The system will allow users to create and interact with content without requiring personal identification or sensitive disclosures.
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Twine Game 3: Adherence to Medication

This game focuses on creating inspiration to take medication. A leaderboard will be used to create friendly competitiveness that will encourage users to continue to take their medications so they may rank higher than their friends. The streak will encourage users to make sure they are consistent with taking their medications so they may gain more points each day.

Functional Requirements:

- 1. **Logging Medicine & Point Accumulation**
 - 1.1. **Medication Logging:** After a user takes their required medication, they may log that they took it with a simple button press.
 - 1.2. **Point System:** Users will be awarded points for logging their medication.
- 2. **Streak Correlated to Daily Logging**
 - 2.1. **Extra Points For Daily:** The user will obtain a streak after three days of logging their medicine; the streak will grow as they log further adherences to their medication.
- 3. **Community Leaderboard**
 - 3.1. **Leaderboard Placement:** Based on the user's points, users will be placed on a leaderboard to see how they compare with their selected friends who are also on the app.
- 4. **Comprehensive Streak:** Medication schedule should be in line with medical professionals opinion, so for users whose schedule isn't daily, streak and points should be assigned in relation to the true schedule.

Non-Functional Requirements:

- 1. **Performance**
 - 1.1. **Leaderboard Refresh:** The leaderboard should manage live data, updating in real time as anyone's points are updated.
 - 1.2. **UI Latency:** All Swaps between current UI elements must take place within 100ms.
 - 2. **Accessibility**
 - 2.1. **Leaderboard Accessibility:** The leaderboard should be available to view anytime, as long as the user has logged their medication at least once.
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Twine Game 4: Exercise Activity Tracker

This game is focused on tracking the user's completed exercises. The goal is to give users a central place to get all of their exercises done in one place. It will allow users to pause and complete their exercises later. Users will choose to play as either a Human or a Zombie, each with its own list of exercises. As a zombie, the user will have to catch the humans to win. As a human, the user will have to escape the zombies to win. To gain speed in the other class, the user will have to complete their exercises in a set period of time. The faster they complete the exercise, the faster they will be in the game.

Functional Requirements:

1. **Combining User Freedom with Targeted Exercises:**
 - 1.1. **Exercise Lists:** At least two lists tied to in-game classes. Each list holds unique exercises geared toward improving HBP health, so users have a choice while still targeting specific health goals.
 - 1.2. **Exercise Duration Control:** To help users of all athletic levels, have selectable choices of easy to hard duration exercises.
2. **User Customization**
 - 2.1. **Avatar Initialization:** At the start of the game, initialize a customizable avatar to be stored so the user can express themselves.
 - 2.2. **Storing Avatar Choice:** Allow users to save their avatar with their chosen exercise list and difficulty level. Additionally, users can make multiple avatars to store different exercise lists as the user changes through their fitness journey.
3. **Built in Sense of Urgency**
 - 3.1. **Recurring Timers:** Timers will be displayed at the top of the screen to give the user a time limit, giving the users a sense of thrill and intensity.
4. **Educational Aspects**
 - 4.1. **Reasoning for each exercise:** Exercises must be accompanied by an explanation of why it is effective for health improvement and its target.

Non-Functional Requirements:

1. **Performance**
 - 1.1. **Latency on fetching created Avatars:** Fetching avatars from local storage must take within 80ms for each avatar.
2. **Accessibility**
 - 2.1. **Accessible Exercises:** 100% of exercises that require equipment, such as dumbbells, benches, etc., must be paired with an equivalent that doesn't require any equipment.

- 2.2. Easy to Learn Exercises:** Exercises must be able to be explained within 150 words or fewer.
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Twine Game 5: Hypertension Dieting

This game is focused on collecting dietary information alongside providing nutritional education based on the DASH⁵ diet plan. The user will have an animal companion acting as a chef to provide feedback when cooking meals. To do so, users must sort through a combination of ingredients commonly recognized to increase/decrease blood pressure to create a recipe.

Functional Requirements

1. Relational Onboarding System

- 1.1. Companion Initialization:** Upon game start, the system shall initialize a “chef” companion as a relational guide.
- 1.2. Goal Presentation:** The System will display a short list of cooking ingredients according to DASH dieting principles for sorting.
- 1.3. Variable Setup:** The system shall initialize a tracker object to store task completion and progress states.

2. Dietary Tracking & Logging

- 2.1. Card Interface:** The interface will provide a clickable list of ingredients organized as cards that are recognized to either increase or decrease blood pressure.
- 2.2. Date Entry:** A manual input screen for users to record a specific date for task completion will be provided.
- 2.3. Progress Visualization:**
 - 2.3.1. Visual Feedback:** Dynamic progress bar correlated to the data values from the diet tracker based on food quality.
 - 2.3.2. Completion State:** Upon completion of a given task, feedback from the animal companion will be provided based on the ingredient’s compliance with DASH standards.

3. Flutter/Firebase Data Handoff

- 3.1. Data Isolation:** User-logged nutrition will be filtered to identify and isolate metrics related to user performance logs for database transmission.
- 3.2. JavaScript Bridge:** Any collected data will be packaged into a JSON payload and transmitted into the Flutter shell.

⁵ [See Reference](#)

Non-Functional Requirements

1. Performance & Reliability

- 1.1. **UI Latency:** All interactions with the interface, specifically checklist toggles, must trigger within 100ms after receiving input.
- 1.2. **Bridge Throughput:** 100% of the collected data must be successfully handed off to the Flutter shell on the first attempt if a connection is present.
- 1.3. **State Persistence:** All of the completed dietary tasks will be properly stored across game sessions, passage navigation, and refreshes with 0% data loss.

2. Usability & Accessibility

- 2.1. **Companion Engagement:** The initialized “chef” companion will be present throughout the session to deliver 100% of dietary feedback or notes.
- 2.2. **First-Time User Experience:** A user with no prior experience should be able to navigate through onboarding in under 5 minutes.
- 2.3. **Visual Inclusivity:**
 - 2.3.1. **Text Sizing:** Web content should allow for resizing up to 200% without any loss in functionality.
 - 2.3.2. **Color Contrast/Usage:** Text and images of text must have a contrast ratio of at least 4:5:1 and will not be used as a sole indicator of task completion. Secondary icons will be present.
- 2.4. **Linguistic Level:** 100% of game dialogue will be compliant with WCAG⁶ standards, ensuring simple language, consistent terminology, short sentence structure, and actionable content.

3. Technical Readability

- 3.1. **Code Transparency:** 100% of macros/code used for any form of state management or game logic will be preceded by a comment describing its function in plain language.
- 3.2. **Auditability:** Any non-technical user should be able to locate and change a goal string (such as ingredients) within the Twine editor in under 5 minutes.
- 3.3. **Passage Metadata:** All passages will be properly labelled with descriptive tags to allow users to search and filter passages in under 10 seconds.

⁶ [See Reference](#)

Environmental Requirements

These are the requirements placed on us by the Playful Health Technology Lab relating to legal and technological requirements. As such, we have identified 3 key requirements for discussion:

1. Right to Design For the Navajo Community

The Playful Health Lab has been given the right by community leaders within the Navajo Community to use designs and imagery that are culturally resonant within the Navajo Community. However, in the name of protection from appropriation for other Indigenous Peoples, the Lab has made it clear that we cannot use any specific designs from other Indigenous Peoples. In the same vein, we will be avoiding using any culturally significant designs that we cannot confirm fall into the category of specifically Navajo culture. For the protection and respect of Indigenous peoples, we will only be using designs we know with certainty have been greenlit for use in our project.

2. Necessity of Using Twine

We have been asked to create all games using Twine. Twine is a simple game engine geared towards enabling non-technical users to create narrative-driven games; simple mini-games are also possible within the engine. Twine is free and open source, so users will be able to freely create and share their own projects without worry. Twine also allows for building complexity by allowing JavaScript code to be integrated into the games, so people who wish to go further with the engine can. This is all in the Lab's vision of cultivating an active and creative community where users can define their own healthcare journey. User-generated content (UGC) is a large category for measuring the success of this project, and Twine was identified by the Lab as being best suited for enabling the creation of UGC.

3. Firebase Server Connections

Firebase is a technology we are required to connect to within our Twine JavaScript functions to enable the telemetry data collection. The Lab has decided to use Firebase so they can use Firestore, a scalable NoSQL cloud database built on Google Cloud. It will sync user data with the Lab's servers, allowing for live data collection. Dr. Duval has made it clear that much of this integration will be handled by our Team lead, Pavan Gorintla. The use of the Firestore functions will be our responsibility within the game's backends. Our only concern is the proper calling of the Functions developed by Pavan in the proper places.

Potential Risks

As a health-based app, the potential risks must be scrutinized heavily. Our analysis took place at a domain level, above the individual games, focusing on identifying the major risks so that each game can be developed with these major risks in mind:

1. **Accidental Data Exposure**
2. **Users are being rewarded for false reporting.**
3. **Moderating User-Generated Content**

We address these risks and discuss how they can affect the stakeholders involved, namely the users and the Playful Health Lab.

1. Accidental Data Exposure

As we will be working with users' health statistics, any accidental data exposure would be detrimental to the trust within the community and to the reputation of the Playful Health Lab. This is exacerbated by our goal of building trust within historically underrepresented communities; a breach of privacy would be among the most disastrous errors to occur. As for how this could occur, we realistically see a couple of ways this could take place.

1. **Targeted “Hacking”:** Users will be protected by a username and password identification. So, the simplest way a user's personal information could be leaked would be the sharing of individuals' passwords and usernames. To avoid this, passwords will never be stored as plain text anywhere within internal databases; interpersonal sharing within the community may allow for bad actors to gather information regardless.
2. **Database Attacks:** In real-world examples, malicious attacks against entities that gather health-related info are a real risk the Lab faces. Immediately, the *23andMe* data leak brings concern over the protection of users who put their personal information into an online database of any kind.

While the defense against Database attacks may fall closer to the purview of Firebase, the technology the Lab will be using to store user data, it is likely an ethically proper idea to disclose these risks to the user to ensure a transparent experience. Further, to prevent targeted hacking, the app should disclose proper safety methods, including keeping their usernames and passwords in a safe place, and making it clear that the Lab will never request their password, and to distrust any entity that asks. To build in a method of protection, we also suggest two-factor authentication to further insulate the user's personal data.

2. Users Being Rewarded for False Reporting

As a user's health statistics will directly impact the games they are playing, we run the risk of creating an environment where users are immediately rewarded for lying about the statistics they report. In doing so, the data being collected could become skewed as users report better health measurements than they actually have in order to achieve better results in the games they play. Furthermore, one of the main goals of the project, to enhance HBP treatment compliance, is put into jeopardy as lying about compliance becomes rewarded. This risk would largely result in issues in the research aspect of this project being conducted by the Playful Health Lab. As these data figures are planned to be used for an R01 application through the NIH Research Project Grant, the legitimacy of their collection will be up to absolute scrutiny. A possible resolution to this risk would be to increase the number of clinical tools that can connect with the app, so user data is collected by an unbiased tool.

3. Moderating User-Generated Content

A risk described by our Client, Dr. Duval, is the idea that the mini-games we will be producing will eventually lose novelty without fresh content being produced. To remedy this, a large goal of our solution is to enable user control and creativity over their own healthcare journey by hosting User Generated Content (UGC) on the platform. However, this requirement itself runs the risk of moderation over this content. At scale, the UGC runs the risk of becoming un-moderatable. As more users join the platform and publish content, the rate at which moderation needs to be held can grow rapidly. The time necessary is also dependent on the depth of the game, as simple games may only take a few minutes, but fleshed-out projects may obscure harmful content behind layers of choices and logic. We have found a few options to be discussed within the Lab:

1. Restricting the ages of users who can navigate the UGC.
2. Creation of punishments for Terms of Service Breaches.
3. Medium to large scaling costs, hiring either employees or developing an automated moderation service.

Risk Summary

This list is non-exhaustive and mainly serves to identify the largest risks involved with this project. Remainder risks may still be identified and reported to the Lab as needed based on relevance and severity. However, it is important to note that all risks are deserving of the same level of analysis as the above list. Our risks were based on industry standards presented in the ACM Code of Ethics, as well as on real-world events that displayed how risks weren't successfully avoided.

Project Schedule

The execution of CardioCare Quest is organized into ten high-level milestones. It's important to note that at this early stage of development, the project schedule is still being finalized. Each milestone listed here is not comprehensive, and work completed towards each milestone is not entirely self-contained from the rest.

Milestones

- **Milestone 1: Storyboarding & Gameplay Flow:** Conceptualize and finalize storyboards for all 5 games during sessions with the PHT Lab to ensure a proper base for the game development phase.
- **Milestone 2: Prototyping:** Build individual prototypes for each of the 5 factors, focusing on establishing the functionality required for each unique gameplay mechanic.
- **Milestone 3: Functional Loop:** Establish the first playable loop for each game to verify proper state management and data handoff functionality.
- **Milestone 4: Bridge Configuration:** Properly establish the Twine-to-Flutter communication bridge to ensure that all hypertension metrics can be sent to the backend database.
- **Milestone 5: Alpha Version:** Tie together all core mechanics and systems required for each game using placeholders or other unpolished assets to create a feature-complete system despite lacking polish.
- **Milestone 6: Beta Version:** Replace all placeholder assets with completed versions of text, audio, images, etc., while also focusing on refining the gameplay functionality and rigorous bug-fixing.
- **Milestone 7: Technical Readability & Documentation Audit:** Perform a comprehensive audit of the entire code base to ensure it is organized, understandable, and well-documented for future community use. This also includes assessing compliance with accessibility standards.
- **Milestone 8: Cross-Platform Audit & Other Testing:** Perform final testing for Android/iOS compatibility and conduct any other required tests for usability, accessibility, and bug finding.
- **Milestone 9: Final Build:** Finalize a “Release Candidate” build, ensuring that there are no major bugs and that game mechanics are fully fleshed out and tested.
- **Milestone 10: App Shell Integration:** Finalize any changes required for the Flutter App Shell, ensuring proper navigation between all sections of the platform. Twine games will also be fully integrated into the platform at this point for usage.

CARDIOCORE QUEST TIMELINE

Task	Start Week	End Week
Storyboarding & Gameplay Flow	Week 14	Week 15
Prototyping	Week 15	Week 16
Functional Loop	Week 16	Week 17
Bridge Configuration	Week 17	Week 18
Alpha Version	Week 18	Week 19
Beta Version	Week 19	Week 20
Accessibility Audit	Week 19	Week 20
Bug Testing	Week 18	Week 20
Final Build	Week 20	Week 21
Integration	Week 16	Week 17
Launch	Week 20	Week 21

Timeline Details:

- Week 14:** Start of Storyboarding & Gameplay Flow.
- Week 15:** End of Storyboarding & Gameplay Flow; Start of Prototyping.
- Week 16:** End of Prototyping; Start of Functional Loop.
- Week 17:** End of Functional Loop; Start of Bridge Configuration.
- Week 18:** End of Bridge Configuration; Start of Alpha Version.
- Week 19:** End of Alpha Version; Start of Beta Version.
- Week 20:** End of Beta Version; Start of Accessibility Audit and Bug Testing.
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- Week 19:** End of Alpha Version; Start of Beta Version.
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- Week 21:** End of Accessibility Audit and Bug Testing; Start of Final Build.

Conclusion

Hypertension remains a critical global health issue that disproportionately affects indigenous communities due to a direct manifestation of systematic technological and healthcare inequalities. But despite having a rapidly growing mobile health industry, existing tools continue to fall short by prioritizing generalized and clinical approaches that fail to resonate with the lived experiences of underserved populations. This disconnect limits user engagement and contributes to ongoing disparities in treatment adherence and health outcomes.

CardioCare Quest directly addresses this gap by proposing a culturally grounded, game-based health platform that challenges the homogenization present in modern health applications. The platform enables users to track hypertension-related data through interactive twine-based games such as education games, medication and exercise activity trackers, and dieting logging experiences. These games establish a strong foundation for medical data collection by embedding self-reporting within engaging, narrative-driven interactions that reflect community values and traditions. By shifting medical data collection into a more culturally meaningful context, CardioCare Quest can transform health tracking from a clinical task into an accessible and empowering experience.

A key insight guiding the future development of CardioCare Quest is the importance of designing systems that do not simply deliver information but instead offer users an opportunity to actively participate in their own health journey. By moving beyond traditional medical applications, the platform offers a more flexible, inclusive, and culturally relevant model of digital health engagement. In the future, we will focus on developing functional prototypes of the discussed games that meet both technical requirements and the needs of the communities they are intended to serve.

In conclusion, CardioCare Quest represents a meaningful step toward bridging the gap between healthcare technology and cultural relevance, creating opportunities to strengthen user engagement, build trust, and reduce disparities within underserved communities.

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